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Economic impact of productivity losses attributable to porcine reproductive and respiratory syndrome virus in United States pork production, 2016–2020

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ABSTRACT

The porcine reproductive and respiratory syndrome virus (PRRSV) continues to pose a major economic burden on the swine industry in the United States (US). This study estimates the annual cost of PRRSV-related productivity losses using US data from 2016 to 2020. Productivity data were collected from commercial swine herds with known PRRSV, porcine epidemic diarrhea virus (PEDV), and *Mycoplasma hyopneumoniae* statuses. The cost of PRRSV was estimated separately for breeding and growing herds. Data from 12 swine companies were used to estimate the cost of PRRSV in the breeding herds; PRRSV statuses were assigned to the breeding herd weeks based on the American Association of Swine Veterinarians (AASV) herd classification and outbreak history as follows: Breeding Herd (BH)-A if PRRSV-negative or provisionally negative, BH-B if PRRSV-positive stable, BH-C if PRRSV-positive unstable after 16-weeks post-outbreak, BH-D if within 16 weeks post-outbreak in PRRSV-affected herds, and BH-E if within 16 weeks post-outbreak in PRRSV-unaffected herds. Data from nine swine companies were used to estimate the cost of PRRSV in the growing herds, and statuses were assigned to growing groups as follows: Growing Herd (GH)-A if PRRSV-negative from weaning to market, GH-B if they were PRRSV-negative at weaning but became PRRSV-positive by market, and GH-C if they were weaned PRRSV-positive. Data on the distribution of US breeding herds by PRRSV status were sourced from the Morrison Swine Health Monitoring Program (MSHMP) database. Swine production-associated prices and costs were sourced from the USDA and InterPIG. An enterprise budgeting model was used to estimate the economic losses from reduced productivity caused by PRRSV. PRRSV caused annual losses of 380.82 million USD in the breeding phase and 819.41 million USD in the growing phase, 1.2 billion USD in total which is a significant increase from the 663.91 million USD reported in 2013. The updated estimates highlight the continuous and growing burden of PRRSV on US swine production.

1. Introduction

The porcine reproductive and respiratory syndrome virus (PRRSV) has remained one of the most challenging and costly diseases affecting the swine industry in the United States (US) despite substantial efforts toward its prevention, control and elimination (Magalhães et al., 2021; Perez et al., 2015; Valdes-Donoso et al., 2016). The virus significantly compromises reproductive and respiratory health in pigs, leading to substantial productivity losses that continue to burden pork producers. Given PRRSV's persistence and the complexities of its control and eradication, a comprehensive understanding of its economic impact is crucial for shaping effective company, regional, and national policies. Such insights are vital for evaluating current management strategies' success, identifying improvement areas, and ensuring that resources are allocated efficiently to mitigate the disease's burden.

Economic analyses have long been essential in quantifying the financial consequences of animal diseases and evaluating control strategies (Dijkhuizen et al., 1995); Methods such as cost-benefit analysis,

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partial budgeting, and cost-effectiveness analysis help determine the financial viability of preventive and control measures (Bester et al., 2024; Goldbach and Alban, 2006; Jerlström et al., 2022; Niemi et al., 2021). Additional tools, including risk analysis, opportunity cost analysis, and input-output models, provide insights into potential risks and lost revenue due to disease (Lindberg and Hansson, 2009; Meuwissen et al., 1999; Van Eenennaam et al., 2021). Enterprise budget modeling has been particularly useful for estimating the economic impact of PRRSV in the US, as it allows for a unified quantification of disease-induced losses by modeling standardized income and expenses across herds (Holtkamp et al., 2013).

Two major studies had previously quantified the annual cost of lost productivity attributable to PRRSV to US swine producers; the first study by Neumann et al. in 2005 estimated this cost at $561 million US dollars (Neumann et al., 2005), and the study by Holtkamp et al. in 2013 updated this cost to about $664 million US dollars, based on data from 2005 to 2010 (Holtkamp et al., 2013).

Since the last estimate was published, the swine industry has experienced some noteworthy events that could have influenced the financial impact of PRRSV. The most notable was the entry of the porcine epidemic diarrhea virus (PEDV) into the US in 2013, which imposed significant economic losses to US pork producers (Huang et al., 2013; Schulz and Tonsor, 2015). Studies have demonstrated synergistic dips in swine productivity from PEDV and PRRSV coinfections (Qiu et al., 2022; Schweer et al., 2016). There have also been more commercially available modified live virus vaccine products (Li et al., 2024), accompanied by the increased use of these vaccines (Kikuti et al., 2021), with more swine producers opting to control PRRSV in their herds rather than eliminate it (Morrison Swine Health Monitoring Project, 2024). From 2010 to the present, the number and size of US swine companies have seen significant changes, with fewer companies now controlling a larger share of the market (https://www.nass.usda.gov/Publications/Highlights/2024/Census22_HL_Hogs_Pigs.pdf, Number of Farms down since 2017, 2024). During this period, the swine industry also experienced increased routine monitoring and surveillance, evidenced by the continued increase in samples sent to the veterinary diagnostic laboratories for PRRSV testing (Trevisan et al., 2019b). The observed increase in monitoring may have stemmed from the industry-wide efforts to standardize terminologies for describing the PRRSV status of swine herds (Holtkamp et al., 2021, 2011) based on laboratory outcomes from routine PRRSV monitoring. The shift from individual animal sampling to the more cost-efficient aggregate sample types may have also contributed to the adoption of best practices for monitoring and surveillance (Almeida et al., 2020; Osemeke et al., 2022; Trevisan et al., 2019a, 2019b; Vilalta et al., 2019). This increased PRRSV monitoring, and the development of high-end molecular techniques significantly improved the understanding of the PRRS virus and its ecology (Trevisan et al., 2022; Zhang et al., 2016). Since the last estimate for the cost of PRRSV was published, the swine industry has also experienced the emergence and reemergence of PRRSV variants, often accompanied by atypical losses (Cheng et al., 2022; Wang et al., 2015).

Given these industry changes and new scientific insights, an updated economic analysis is essential to reflect the current cost of PRRSV to the US swine industry. This study aims to provide an updated estimate of the annual economic impact of PRRSV on the US swine industry between 2016 and 2020 and identify the key factors driving any shifts in the financial burden of this virus.

2. Materials and methods

2.1. Overview

This study estimates the annual cost of lost productivity in US swine herds attributable to PRRSV from 2016 to 2020. To enable comparisons over time, the approach adopted in this study was similar to that used by Holtkamp et al. (2013), with the economic impact of PRRSV estimated separately for breeding and growing pig herds. The factors evaluated that contributed to the economic impact in the US included: the proportion of swine herds in the US affected (American Association of Swine Veterinarians [AASV] IA, IB, II, and IIvx) and unaffected (AASV III and IV) categories during the study period; the productivity differences between PRRSV-affected- and unaffected herds; the prices and costs of inputs and outputs; and the national inventories of domestic and imported swine. For breeding-herd estimates, sow performance records, disease outbreak records, vaccination records, and the disease statuses by week for PRRSV, porcine epidemic diarrhea virus (PEDV), and *Mycoplasma hyopneumoniae* (*Mhp*) for the study period were obtained from 12 participating swine companies. Data from nine swine companies collected through farm records and surveys were used to estimate the cost of PRRSV in the growing herd. The distribution of US swine breeding herds in each PRRSV category throughout the study period was extrapolated from estimates of PRRSV occurrence from the Morrison Swine Health Monitoring Program (MSHMP) at the University of Minnesota College of Veterinary Medicine. Prices and costs were obtained from the United States Department of Agriculture and InterPig websites. The productivity measures assessed and data from the US herd distribution by PRRSV status were entered into an enterprise budgeting model to estimate the economic value of lost productivity attributable to PRRSV.

2.2. Herd classification

For the breeding herds, PRRSV statuses were assigned based on the American Association of Swine Veterinarians (AASV) PRRSV classification for breeding herds (Holtkamp et al., 2021) and outbreak history (Fig. 1, Table S2). An outbreak week was defined as a herd week deemed so by the attending veterinarian of that herd. An outbreak week would generally have laboratory evidence of PRRSV in a population of pigs that

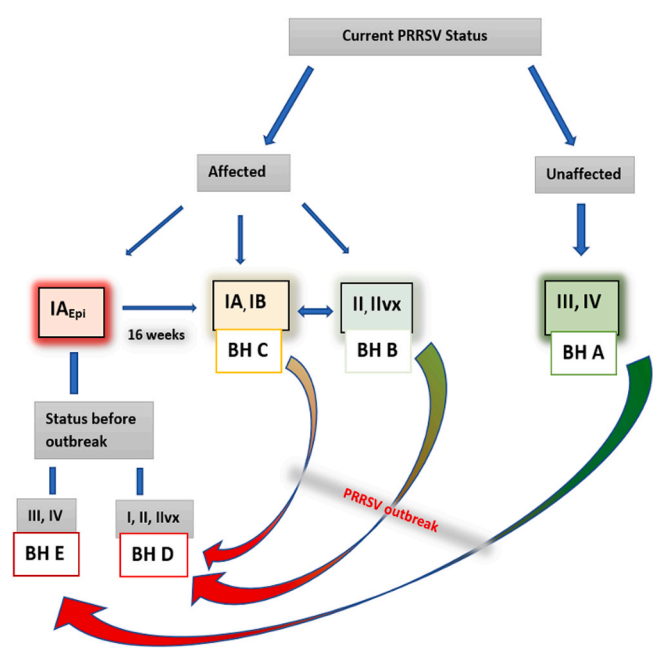

Fig. 1. Classification of breeding herds for estimating productivity losses attributable to PRRSV. BH-A is assigned to PRRSV-negative or provisionally negative (American Association of Swine Veterinarians [AASV] categories III and IV) herd weeks, BH-B to PRRSV-positive stable herd weeks (AASV categories II or IIvx), BH-C to PRRSV-positive unstable herd weeks 16-weeks or more post-outbreak (AASV categories IA or IB), BH-D to PRRSV-positive unstable herd weeks within 16 weeks of an outbreak and PRRSV-positive unstable or stable before the outbreak (AASV category IA), and BH-E to PRRSV-positive unstable herd weeks within 16 weeks of an outbreak and PRRSV-provisional negative or negative before the outbreak.

was previously PRRSV-unaffected (AASV categories III and IV) or the laboratory confirmation of the entry of a new PRRSV variant into a previously PRRSV-positive (AASV statuses IA, IB, II, and IIvx) swine population with accompanying clinical signs and productivity dips. Breeding-herd weeks were assigned statuses Breeding herd-A or BH-A if they were PRRSV-negative or provisionally negative (AASV categories III and IV), BH-B if PRRSV-positive stable (AASV categories II or IIvx), BH-C if PRRSV-positive unstable 16-weeks or more post-outbreak (AASV categories IA or IB), BH-D if within 16 weeks post-outbreak and PRRSV-positive before the outbreak (AASV category IA), and BH-E if within 16 weeks post-outbreak and PRRSV-provisional negative or negative before the outbreak (AASV category IA). The number of herd-weeks in each PRRSV category per year was calculated, and these estimates were extrapolated to all US breeding herds.

For the growing pigs, if a group was PRRSV-wildtype negative when they were weaned and stayed negative until market, that group was designated Growing Herd-A or GH-A; if the group was PRRSV-wildtype negative at weaning but had laboratory evidence of PRRSV-wildtype detection any time before market, that group was designated GH-B. If the group was PRRSV-wildtype positive at weaning, that group was designated GH-C (Fig. 2, Table S2).

To account for other disease pathogens and coinfections that could significantly affect the productivity parameters being measured, breeding herd weeks were categorized for *Mhp* as follows: A (negative), B (controlled or non-clinical), and C (uncontrolled or clinical). PEDV was classified as A (positive-stable, provisionally negative, or negative) and B (positive-unstable). The classification systems for *Mhp* and PEDV were chosen as they presented the most practical method to unify the different herd classification systems used by the participating swine companies.

2.3. Herd distribution

The distribution of US breeding females and growing pigs was sourced from the MSHMP database (MSHMP; https://mshmp.umn.edu/reports/). Established in 2011, the MSHMP houses data for over 60 % of the US breeding herds, encompassing more than 3.5 million sows. The project collects weekly data on herd health and estimates the cumulative incidence and prevalence of PRRSV, PEDV, and other significant swine diseases. Annual proportions of the herd weeks that were within each of the breeding herd categories were obtained. Similarly, the proportion of breeding herds in the PRRSV-positive unstable categories (AASV categories IA and IB) per year was used to estimate the proportion of PRRSV-positive pigs being weaned in the US. By definition, AASV categories IA and IB herds wean pigs that are positive for PRRSV. The data obtained from a survey (S3 Table) of swine

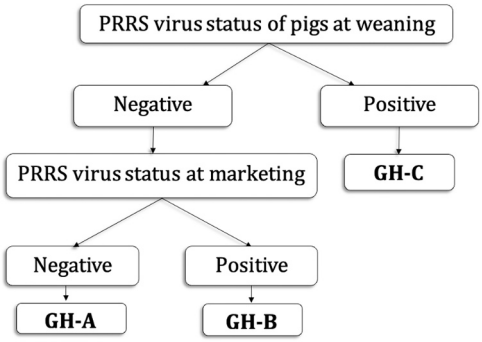

Fig. 2. Classification of growing herds for estimating productivity losses attributable to PRRSV. GH-A groups are wean-to-market groups weaned negative and stayed negative to market, GH-B groups are wean-to-market groups weaned negative but tested PRRSV wild-type positive at some point before market, GH-C groups are wean-to-market groups that are weaned PRRSV wildtype positive.

veterinarians was used to estimate the proportion of pigs that were weaned or placed PRRSV-negative and stayed PRRSV-negative until market.

2.4. Productivity losses attributable to PRRSV

2.4.1. Breeding-herd productivity

Weekly records of the liveborn per litter (LBL), the preweaning mortality rate (PWM), litters per mated female per year (LMFY), sow culling rates (SCR), and sow mortality rates (SMR) were obtained from US breeding herds with known PRRSV, PEDV, and *Mhp* statuses for the study period. A total of 297 breed-to-wean farms across 12 participating production systems were included in this study, representing a combined female inventory of 1,112,809 sows, accounting for about 17.82 % of the US sow inventory during the study period. Over the study period, there were 68,800 analyzable herd weeks, with an average mated female inventory of 3759.49 (ranging from 224.71 to 10,794.69). These swine companies used different record-keeping systems to house their data, namely, PigCHAMP, MetaFarms, and farm-owned computer systems. The participating swine companies were a convenient sample of US companies willing to participate in the study and had records spanning the study period. Most of the farms belonging to these swine companies were located in the Midwest (the primary swine production region of the US), with a few extending into the South and Northeastern regions of the US.

The pigs weaned per mated sows per year was estimated using the LBL, LMFY, and PWM data from the production records, and the breeding female replacement rates were estimated using the sow mortality and culling rates.

2.4.2. Growing herds productivity

Data for growing pig herd was obtained from two sources: farm records and a survey. The first source was productivity and disease records of wean-to-finish groups spanning the study period. The average daily gains (ADG), feed conversion ratio (FCR), mortality rates (MOR), and percentage of pigs sold in the primary market (PPPM) were obtained for wean-to-finish groups of pigs. The swine-pathogen records requested were for PRRSV, PEDV, and *Mhp*. For groups that had separate nursery and finisher production phases, the records were also included if the integrity of the groups was maintained when transferred from the nursery to the finishing phase, and the swine company had an internal system for summarizing the productivity and disease data into a single wean-to-finish close-out record. Groups having diagnostic evidence of wild-type PRRSV infection pre- and post-placement in grow-finish spaces were included. Groups that did not have or share sequencing records to confirm that the PRRSV variant in positive RT-qPCR results was wild-type, to facilitate the classification of such herds as GH-A or GH-B, were excluded. Consequently, data from seven (out of eight) production systems were excluded from the analysis. The second source of data for the growing pig herd was from survey questionnaires sent to veterinarians to obtain estimates of the proportion of the growing pigs within the companies they offered services to for the study period. Productivity (ADG, FCR, percentage mortality, and PPPM) estimates, treatment costs, vaccination costs, diagnostic costs, and the annual number of pigs marketed for each PRRSV category were also requested in the surveys. The survey encompassed swine herds of different sizes across different regions. Surveys (Supplementary file S1) were distributed to 26 veterinarians; 10 responded, but only eight provided analyzable data. One response contained numerous missing fields, and another was excluded because of excessive errors. Veterinarians shared insights based on company-specific data on productivity and disease records of wean-to-finish groups and their direct experience with the herds they serviced, with precautions taken to avoid duplicated data from herds overseen by multiple veterinarians. Each veterinarian completed the survey for a single company and when a company had multiple veterinarians, only one was allowed to complete the survey to

avoid duplication. Altogether, productivity data from one swine company and survey data from eight companies were used for this analysis, totaling nine swine companies. The company with eligible productivity data markets about 850,000 pigs annually and had the required productivity and diagnostic information for 272 wean-to-market groups within the study period. The annual number of pigs marketed for the companies participating in the survey ranged from 680,000 to 5,800,000. Taken together, the nine swine companies marketed about 20,030,000 pigs annually, accounting for about 16 % of all US pigs marketed during the study period. Of the nine swine companies, seven had all grow-finish sites in the Midwest, one operated exclusively in the South, and one maintained sites in both the Midwest and the West.

2.4.3. Statistical analysis of herd performance

To estimate the least-squares mean productivity for each PRRSV status in the breeding and growing pig herds, linear mixed regression models were used. The unit of analysis for productivity measures was the herd-week for breeding herd and the group (wean to finish closeout) for growing herds. For the breeding herd, separate models were built for each productivity measure (LBL, LMFY, PWM, SCR, SMR), with PRRSV status, PEDV status, *Mhp* status, and the interaction between PRRSV and *Mhp* included as fixed effects to control for disease infections and co-infections. Random effects accounted for farm nested within the production system, the production system, and season nested within each year. Herd weeks that coincided with herd depopulation events were excluded. In addition, sow-performance outliers falling outside biologically plausible thresholds as identified by the study team were removed before analysis. A log transformation was applied when the assumptions of homoscedasticity (constant variance) or normality of residuals were violated, consequently, the PWM, SCR, and SMR were log-transformed before modeling, correcting deviations from those assumptions, and back-transformed to obtain least-squares means in the original units. For the growing pig herd data, similar to the breeding herd, separate models were used for each productivity measure (ADG, FCR, MOR, PPPM), with PRRSV status, PEDV status, *Mhp* status included as fixed effects. The farm (site ID) and the seasons nested within a year, and year were used as random effects. All productivity estimates from both sources (data and survey) were averaged using the annual number of pigs marketed from the swine companies as weights. All statistical analyses were conducted in JMP statistical software (SAS Institute Inc., 2024).

2.5. Prices, costs, swine inventories, and the economic model

An enterprise budgeting model, similar to that used in the 2013 PRRSV economics study (Holtkamp et al., 2013), was utilized to estimate the total economic value of productivity losses due to PRRSV in the US swine industry. The model was developed in Microsoft Excel (Microsoft Corporation, Redmond, Washington) and is available upon request from the corresponding author for educational purposes and non-commercial research. The model allowed for the evaluation of alternative scenarios with varying levels of productivity while using standard reference values for market pig prices and input costs to isolate the impact of the varying levels of productivity on profitability.

The model was applied separately to breeding herds with scenarios included for each of the five PRRSV classifications. The least-squares mean productivity estimates for each PRRSV classification were used to reflect the varying productivity levels for each scenario in the enterprise budgeting model, regardless of the statistical significance of pairwise differences. While some differences in the least-squares mean productivity estimates between PRRSV categories were not statistically significant ($p > 0.05$), these estimates were plugged in the economic model regardless. These differences, however subtle, when combined with other economic factors or scaled to the national level, could still have meaningful financial implications. A similar approach was used for growing pig herds with scenarios for all three PRRSV classifications, assuming a fixed time for pigs to reach market weight.

Standard reference values for fixed costs, input prices, weaned-pig prices, market-hog prices, and other non-PRRS-affected production metrics were sourced from the USDA (USDA; https://www.usda.gov/) and InterPig (Holtkamp et al., 2022) reports. The weaned-pig price ($37.22 per pig) and market-hog price ($67.8 per cwt of carcass weight) were averages from 2016 to 2020. Breeding-herd and growing-herd feed costs were based on InterPig data and USDA prices of feed components. The breeding-herd diet cost was calculated at $177.28 per ton ($195.44 per metric ton), and for growing pigs, $183.08 per ton ($201.74 per metric ton).

Non-feed variable costs in the breeding-herd model included breeding expenses, animal health, labor, and other miscellaneous costs (e.g., utilities, supplies). The growing-herd model mirrored these categories, except for breeding expenses. The weaned pig price used in the breeding-herd model to calculate revenue and in the growing pig model to calculate the cost of weaned pigs was the same. Fixed costs in both models included depreciation, interest, repairs, taxes, and insurance for assets such as buildings and equipment. Depreciation was calculated using the straight-line method (Archibald, 1967), while the other fixed costs were estimated as percentages of capital investments. Parameter values used for non-feed variable and fixed costs, as well as production metrics unaffected by PRRSV, are presented in Tables 1 and 2.

The annual cost of lost productivity attributable to PRRSV in the US was calculated as the difference in net profit between two scenarios: one where the US swine industry operates as entirely PRRSV-unaffected, and the other reflecting the current situation, where part of the swine population is PRRSV-affected. The estimates from the separate economic analysis for the breeding and growing pig herds were combined to obtain the total annual cost of productivity losses attributable to PRRSV.

For the breeding-herd analysis, a scenario where all the sows in the US are negative for PRRSV (or in the BH-A category) was created and the alternative scenario was also created, representing the current reality of the US swine industry with all five PRRSV categories (BH-A to E) included. Production averages were weighted by the estimated percentage of breeding herds in each PRRSV category for the study period. The weighting factor (percentage of breeding herd weeks in each category) was obtained from the MSHMP database (herd distribution) from 2016 to 2020. Similarly, for the growing-herd analysis, the PRRSV-free scenario was created where all the growing pig herds in the US were PRRSV-unaffected (i.e., classified as GH-A). In contrast, the alternative scenario represented current industry realities and included all three PRRSV categories (GH-A, B, and C), with productivity measures weighted by the proportion of groups in each category. The proportion of sow herds from the MSHMP database that were PRRSV-positive-unstable (AASV statuses 1 A and 1B) during the study period was used as a proxy for the proportion of pigs being weaned positive (GH-C) for the study period; the remaining proportion of weaned pigs (i.e., PRRSV-negative) were divided into GH-A (i.e., PRRSV-negative pigs that stay negative until market) or GH-B (i.e., PRRSV-negative pigs that become PRRSV-infected before market) using the veterinarians' survey data earlier mentioned.

The total cost of PRRSV was scaled to the national level using data on the US breeding-herd inventory, the number of pigs marketed annually, and the number of pigs imported for growing. The USDA estimate of the average breeding-female inventory for 2016–2020 was 6,243,685, and an average of 125,166,200 pigs were marketed annually, including domestic production and imported pigs. On average, 3,397,670 pigs weighing less than 15 lbs (or 6.8 kg) were imported annually for feeding and harvest.

The economic model's predicted number of weaned pigs was adjusted to match the market figures reported by the USDA by using the "Goal Seek" function in Microsoft Excel (Microsoft Corporation, Redmond, Washington) to multiplicatively adjust the LBL and LMFY variables by a factor of 0.964, and PWM by a factor of 1.036.

Table 1
Values used to estimate fixed costs, non-feed variable costs, and other production metrics that are assumed to be unaffected by porcine reproductive and respiratory syndrome virus (PRRSV) in the breeding herds.

Category	Subcategory	Metric	Value
Performance	-	Average lactation length (days)	20.00
		Average gestation length (days)	115.00
		Average weaning weight (lb/pig)	12.00 (or 5.44 kg)
Non-Feed Variable Costs	Animal Health	Cost per breeding female per year ($)	17.50
		Cost per pig weaned ($)	0.75
	Breeding	Cost per breeding female per year ($)	73.80
	Labor and Farm Management	Labor and management hours per breeding female per year	8.24
		Wages and benefits ($ per hour)	13.98
	Other Non-Feed Variable Costs	Cost per breeding female per year ($)	34.20
Fixed Costs	Depreciation on Buildings and Improvements	Capital investment ($ per breeding female)	919.00
		Expected* useful life (years)	25.00
		Salvage# value after useful life (% of capital investment)	10.00
	Depreciation on Machinery and Equipment	Capital investment ($ per breeding female)	563.00
		Expected* useful life (years)	15.00
		Salvage# value after useful life (% of capital investment)	5.00
	Insurance, Repairs, Taxes, and Interest	For breeding animals (% of capital investment per year)	6.00
		For buildings and improvements (% of capital investment per year)	7.00
		For machinery and equipment (% of capital investment per year)	8.00

* Expected useful life is the period an asset is anticipated to remain functional for its primary intended purpose without requiring major repairs # Salvage value is the estimated value of the asset at the end of its useful life.

3. Results

3.1. Herd distribution

The MSHMP database had 227,858 herd weeks from breed-to-wean herds across all five study years (2016–2020) with PRRSV status data. The proportions of herd weeks in each PRRSV status during the study period are reported in Table 3. For the growing pig herds, the proportion of groups of pigs that were PRRSV-positive at placement (GH-C) was 25.76 %. From the survey data, the proportion of all groups placed negative that stayed negative until market was 59 %. Applying this to the proportion of pigs that were PRRSV-negative at placement (74.24 %) from the MSHMP database, the proportion of GH-A groups for the study period was estimated to be 43.80 % and the proportion of GH-B groups was estimated at 30.44 %.

Table 2
Values used to estimate fixed costs, non-feed variable costs, and other production metrics that are assumed to be unaffected by porcine reproductive and respiratory syndrome virus (PRRSV) in the growing herds.

Category	Subcategory	Metric	Value
Growth Performance	-	Carcass yield (%)	75.00
		Average live weight of light-weight pig at market (lb./pig)	157.50 (or 71.44 kg)
Production Times	-	Target days on feed, wean-to-market	167.00
		Average days in nursery	45.00
		Average downtime between turns (days)	2.00
Revenue: Market Pigs	-	Market price of light-weight pigs (% of full value price)	70.00
Variable Costs: Feed	-	Gain at time of death for mortality (% of total to market)	33.00
Variable Costs: Non-Feed	Animal Health	Animal health cost wean-to-finish ($/pig placed)	3.41
	Labor and Farm Management	Hours required per pig placed	0.22
		Wages and benefits ($/hour)	13.98
	Other Non-Feed Variable Costs	Other wean-to-finish non-feed variable costs ($/lb. of gain)	0.02 (or 0.04 $/kg)
Fixed Costs	Depreciation on Buildings and Improvements	Capital investment in nursery buildings and improvements ($/pig space)	89.93
		Capital investment in finisher buildings and improvements ($/pig space)	139.61
		Expected* useful life of buildings and improvements (years)	25.00
		Salvage# value after useful life (% of capital investment)	20.00
	Depreciation on Machinery and Equipment	Capital investment in nursery machinery and equipment ($/pig space)	55.06
		Capital investment in finisher machinery and equipment ($/pig space)	11.28
		Expected* useful life of machinery and equipment (years)	15.00
		Salvage# value after useful life (% of capital investment)	10.00
	Insurance, Repairs, Taxes, and Interest	For buildings and improvements (% of capital investment/year)	7.00
		For machinery and equipment (% of capital investment/year)	10.00

* Expected useful life is the period an asset is anticipated to remain functional for its primary intended purpose without requiring major repairs # Salvage value is the estimated value of the asset at the end of its useful life.

3.2. Productivity differences

3.2.1. Breeding herds

PRRSV status was significantly associated with productivity (Table S3). The five breeding herd categories (BH-A through BH-E) exhibited statistically different ($p < 0.05$) numbers of live born piglets per litter (Table 4), ranging from 13.57 in BH-A to 10.53 in BH-E. Pre-weaning mortality varied significantly as well, ranging from 16.57 % in BH-A to 33.84 % in BH-E, with all categories except BH-C and BH-D

Table 3
The distribution of US breeding herd weeks by PRRSV status from 2016–2020 using the Morrison Swine Health Monitoring Program database. BH-A is assigned to PRRSV-negative or provisionally negative (American Association of Swine Veterinarians [AASV] categories III and IV) herd weeks, BH-B to PRRSV-positive stable herd weeks (AASV categories II or IIvx), BH-C to PRRSV-positive unstable herd weeks 16-weeks or more post-outbreak (AASV categories IA or IB), BH-D to PRRSV-positive unstable herd weeks within 16 weeks of an outbreak and PRRSV-positive unstable or stable before the outbreak (AASV category IA), and BH-E to PRRSV-positive unstable herd weeks within 16 weeks of an outbreak and PRRSV-provisional negative or negative before the outbreak.

Year	Number of Herd-Weeks	BH-A (%)	BH-B (%)	BH-C (%)	BH-D (%)	BH-E (%)
2016	41,847	38.67	37.56	14.90	7.34	1.53
2017	44,600	39.48	37.27	13.94	8.30	1.02
2018	46,166	39.15	33.04	17.86	8.58	1.38
2019	47,336	40.02	35.61	17.53	6.15	0.68
2020	47,909	40.17	30.23	21.20	7.30	1.10
Averages	45,572	39.50	34.74	17.09	7.53	1.14

Table 4
The least squares mean estimates of each breeding herd productivity parameter for each PRRSV category. For each productivity parameter, PRRSV categories with different significance letters are statistically different ($p < 0.05$).

Productivity parameters	PRRSV Category	**Estimate (95 % CI)**	**Significance**
Live born per litter[¥]	BH-A	13.57 (13.17, 13.97)	A
	BH-B	13.17 (12.77, 13.57)	B
	BH-C	12.72 (12.32, 13.12)	C
	BH-D	11.80 (11.38, 12.21)	D
	BH-E	10.53 (10.11, 10.95)	E
Preweaning mortality*	BH-A	16.57 % (15.09 %, 18.20 %)	A
	BH-B	18.13 % (16.52 %, 19.91 %)	B
	BH-C	20.75 % (18.90 %, 22.78 %)	C
	BH-D	22.30 % (20.11 %, 24.73 %)	C
	BH-E	33.84 % (30.42 %, 37.64 %)	D
Litters per mated female per year[þ]	BH-A	2.38 (2.33, 2.42)	A
	BH-B	2.36 (2.31, 2.40)	A
	BH-C	2.28 (2.23, 2.32)	B
	BH-D	2.23 (2.17, 2.29)	B
	BH-E	1.97 (1.90, 2.03)	C
Sow mortality rate[#]	BH-A	10.36 % (6.84 %, 15.69 %)	A
	BH-B	11.47 % (7.57 %, 17.36 %)	B
	BH-C	12.16 % (8.03 %, 18.41 %)	BC
	BH-D	12.01 % (7.88 %, 18.29 %)	C
	BH-E	12.92 % (8.48 %, 19.70 %)	BC
Sow culling rate[#]	BH-A	41.80 % (25.48 %, 68.59 %)	A
	BH-B	38.75 % (23.62 %, 63.58 %)	B
	BH-C	37.67 % (22.96 %, 61.82 %)	B
	BH-D	42.74 % (25.85 %, 70.65 %)	AB
	BH-E	42.11 % (25.31 %, 70.05 %)	AB

¥ the total number of liveborn piglets divided by the total number of litters.
þ Calculated by dividing the total number of litters by the total number of mated females.
* Percentage of pigs born alive that were not weaned
# Percentage of inventoried females that died (SMR) or were culled (SCR) annually

differing statistically ($p < 0.05$). Litters per mated female per year ranged from 2.38 (BH-A) to 1.97 (BH-E), with BH-A and BH-B and BH-C and BH-D pairwise comparisons not differing significantly ($p > 0.05$).

Sow mortality rates ranged from 10.36 % in BH-A to 12.92 % in BH-E, with BH-A differing significantly from all other categories ($p < 0.05$). There was also a significant difference between BH-B and BH-D ($p < 0.05$), but no other mortality rate comparisons were statistically different. The sow culling rate ranged from 37.67 % in BH-C to 42.74 % in BH-D, with BH-A's culling rate differing significantly from BH-B and BH-C ($p < 0.05$), though all other comparisons were not statistically different ($p > 0.05$).

3.2.2. Growing herds

The statistical significance of each model predictor is presented in Tables S4 and S5. Mortality rates were lowest in GH-A (7.02 %) and highest in GH-C (12.23 %), with only the GH-A vs. GH-C comparison being statistically significant ($p < 0.05$). Average daily gains were 1.63, 1.57, and 1.54 for GH-A, GH-B, and GH-C, respectively, with no significant pairwise differences (Table 5). Feed conversion ratios for GH-A, GH-B, and GH-C (marginal mean values of 2.40, 2.42, and 2.45) were not statistically different from each other. Similarly, the percentages of pigs sold in the primary market (92.6 %, 91.2 %, and 89.3 %) showed no statistically significant differences ($p > 0.05$).

3.3. Economic cost of productivity losses

PRRSV reduced the number of pigs weaned per female per year by 1.69 pigs (Table 6). At the national level, this represents a staggering loss of 10.6 million USD weaned pigs annually. The economic toll of PRRSV on the US breeding herd alone attributable to lost productivity, was estimated at $380.82 million USD per year. This translates to a cost of $60.99 per breeding female in the US, the cost comes almost entirely from lost revenue. The PRRSV-unaffected scenario, which assumes all breeding herds are BH-A, had an annual revenue of $392.82 million dollars higher than the PRRSV scenario in the US within the study period.

At the national level, PRRSV reduced the number of pigs marketed

Table 5
The least squares mean estimates of each growing herd productivity parameter for each PRRSV category. For each productivity parameter, PRRSV categories with different significance letters are statistically different ($p < 0.05$).

Productivity parameter	PRRSV Category	Mean	Standard error	Significance
Mortality rate(%)*	GH-A	7.02	0.73	A
	GH-B	9.27	1.13	AB
	GH-C	12.23	0.88	B
Average daily gains (lb per day)[#]	GH-A	1.63 (or 0.74 kg)	0.05	A
	GH-B	1.57 (or 0.71 kg)	0.04	A
	GH-C	1.54 (or 0.70 kg)	0.08	A
Feed conversion ratio[þ]	GH-A	2.66	0.05	A
	GH-B	2.68	0.04	A
	GH-C	2.72	0.06	A
Percentage of pigs sold in primary market[¥]	GH-A	92.60	2.38	A
	GH-B	91.20	1.88	A
	GH-C	89.30	3.10	A

* Calculated by dividing the number of pigs that died by the number of pigs placed at the start, then multiplying by 100.
# Computed as (Final weight−Initial weight)/Days on feed.
þ obtained by dividing the total feed consumed by the total weight gained (*Total feed intake / Total weight gain*).
¥ Determined by taking the number of pigs that reached market weight and were sold through the primary sales channel, divided by the total finishing inventory, and multiplying by 100.

Table 6
Productivity and economic losses attributable to the porcine reproductive and respiratory syndrome virus (PRRSV) in US swine breeding herds from 2016 to 2020.

Category	**Measure**	Assumes all herds are BH-A	Actual status (weighted average of all BH categories)	Difference (Actual status minus all herds are BHA)
US Breeding Female Inventory	Breeding Female Inventory	6,243,685	6,243,685	0
	Litters Farrowed (litters/year)	13,239,516	13,014,710	−224,806
	Pigs Born Alive (pigs/year)	173,218,193	164,577,989	−8640,204
	Pigs Weaned (pigs/year)	144,774,757	134,220,665	−10,554,092
	Pigs Weaned per Female per Year	23.19	21.5	−1.69
Economic Impact (Annual)	Total Revenue ($/year)	$5,388,516,469	$4,995,693,166	-$392,823,303
	Total Fixed Costs ($/year)	$665,295,855	$665,295,855	$0
	Total Variable Costs ($/year)	$3,013,516,672	$3,001,516,114	-$12,000,558
	Total Costs ($/year)	$3,678,812,527	$3,666,811,969	-$12,000,558
	Net ($/year)	$1,709,703,942	$1,328,881,197	-$380,822,745
Economic Impact (Per Breeding Female)	Total Revenue ($/breeding female)	$863.03	$800.12	-$62.92
	Total Fixed Costs ($/breeding female)	$106.56	$106.56	$0.00
	Total Variable Costs ($/breeding female)	$482.65	$480.73	-$1.92
	Total Costs ($/breeding female)	$589.21	$587.28	-$1.92
	Net ($/breeding female)	**$273.83**	**$212.84**	**-$60.99**

per year by 12.60 million pigs or 3.45 billion pounds (1.56 million metric tons) of pork at carcass weight. Consequently, the primary driver of losses in the growing pig herds was reduced revenue, totaling $2.29 billion, due to lower numbers of pigs and pounds of pork produced. Interestingly, production costs were reduced by $1.47 billion, as fewer resources were required to raise the reduced number of pigs to market. The annual productivity losses attributable to PRRSV in US growing pig herds was estimated at $819.41 million (Table 7). This cost translates to $131 per breeding female in the US, 1.56 cents per pound (or 3.44 cents per kg) live weight, or 2.08 cents per pound (or 4.59 cents per kg) of carcass weight.

The total economic value of PRRSV-induced productivity losses across the entire US swine industry was estimated at $1.2 billion per year, equating to a staggering $3.3 million in daily losses. On a per-female basis, this represents an annual cost of $192.23 for every sow in the US breeding-female inventory.

4. Discussion

This study presents updated estimates of the economic impact of the

Table 7
Productivity and economic losses attributable to the porcine reproductive and respiratory syndrome virus (PRRSV) in US swine growing pig herds from 2016 to 2020.

Category	**Measure**	Assumes All Herds Are BH-A and GH-A	Actual status (weighted average of all categories)	**Difference** (actual status minus all herds are BHA and GHA)
Economic impact total	Pigs marketed per year in US	137,769,241	125,166,200	−12,603,041
	Live weight marketed in category (lbs/year)	36,352,640,122 (16,490,696,310 kg/year)	31,755,270,469 (14,407,923,964 kg/year)	−4,597,369,653 (−2,085,964,370 kg/year)
	Carcass weight marketed in category (lbs/year)	27,264,480,091 (12,373,432,167 kg/year)	23,816,452,852 (10,793,452,170 kg/year)	−3,448,027,240 (−1,564,027,240 kg/year)
	Total revenue ($/year)	19,222,079,389	16,934,252,187	−2,287,827,202
	Total weaned pig costs ($/year)	5,514,977,739	5,122,154,436	−392,823,303
	Total fixed costs ($/year)	1,111,195,755	1,032,047,006	−79,148,749
	Total variable costs ($/year)	10,032,399,366	9,035,959,081	−996,440,284
	Total costs ($/year)	16,658,572,860	15,190,160,523	−1,468,412,336
	Net ($/year)	**2,563,506,529**	**1,744,091,664**	**−819,414,866**
Economic Impact (Per pig marketed)	Total revenue ($/pig marketed)	$139.52	$135.29	-$4.23
	Total weaned pig costs ($/pig marketed)	40.03	40.92	0.89
	Total fixed costs ($/pig marketed)	8.07	8.25	0.18
	Total variable costs ($/pig marketed)	72.82	72.19	−0.63
	Total costs ($/pig marketed)	120.92	121.36	0.44
	Net ($/pig marketed)	18.61	14	−4.67
Economic Impact (Per breeding female)	Total revenue ($/breeding female)	3078.64	2712.22	−366.42
	Total weaned pig costs ($/breeding female)	883.29	820.37	−62.92
	Total fixed costs ($/breeding female)	177.97	165.29	−12.68
	Total variable costs ($/breeding female)	1606.81	1447.22	−159.59
	Total costs ($/breeding female)	2668.07	2432.88	−235.18
	Net ($/breeding female)	410.58	279.34	−131.24

PRRSV on the US swine industry, focusing on data from 2016 to 2020. By following a similar approach used by Holtkamp et al. (2013), we aimed to create comparable economic estimates to allow for a meaningful examination of how the impact of PRRSV on productivity and profitability has shifted over time. Certain improvements in available data sources since the last estimate helped incorporate subtle nuances into the methodology of this study. Firstly, the growing adoption of the AASV surveillance-based herd classification for PRRSV (Holtkamp et al., 2021, 2011) afforded this study relatively more granularity in the PRRSV classification of breeding herds. For the same reason, there was also more standardization in the PRRSV classification systems across companies, enabling ease in consolidating data across systems with reasonable accuracy. Secondly, the herd distribution estimates for this study were obtained from the MSHMP database, which has also seen increased herd enrolments over time, and currently has a database boasting data from most swine companies in the US and representing over 3.5 million sows. The size and diversity of herds represented further strengthen the validity of our findings and give stakeholders a clearer picture of how the distribution of PRRSV-affected and PRRSV-unaffected breeding herds by PRRSV has changed over time in the US swine industry.

Over the five-year study period, on average, 39.5 % of the breeding herd weeks were classified as BH-A (PRRSV-negative or provisionally negative), an increase in proportion compared to the same estimate of 28 percent from the previous study published in 2013 (Holtkamp et al., 2013). It is possible that some of the observed changes are an artifact of the different data sources; while this study obtained herd distribution data from a PRRSV surveillance database (MSHMP), the previous study sourced data from a survey of veterinarians. The earliest period for which the MSHMP PRRSV surveillance database has publicly available information on PRRSV status distribution of US breeding herd is July of 2009; consequently, there is insufficient information to make a valid comparison of the proportion of BH-A herds in this study and the previous. The classification scheme for PRRSV-affected breeding herds used in this study is different from the classification used in the 2013 study, as this study captures the increased classification granularity afforded by the growing adoption of the AASV herd classification for PRRSV, consequently, valid comparisons cannot be made. Nonetheless, using the AASV classification scheme, it can be noted that 37.7 % of breeding herds weeks were PRRSV-positive stable, with 25.8 % of all breeding herd weeks PRRSV-positive unstable during the study period. A previous study highlighted an increasing trend in US herds classified as PRRSV-positive unstable, attributing this rise to the widespread adoption of a more reliable and effective sample type for PRRSV surveillance (Kikuti et al., 2022). In the previous study, 23 % of the growing pig groups were classified as GH-A (groups that were placed negative and stayed negative until market), 38 % as GH-B (groups that were placed negative but tested positive while on-feed), and 39 % were GP-C (groups that were placed positive); in this study, these proportions were 43.80 % for GH-A, 30.44 % for GH-B, and 25.76 % for GH-C. The PRRSV classification scheme for growing herds in this study was the same as that used in the 2013 study. However, the changes from the 2013 publication should be interpreted with caution since the source of the data in the current study was the MSHMP and a survey of veterinarians, whereas the source for the study published in 2013 was a survey of veterinarians and data from producers.

For the breeding phase of swine production, the 2013 study estimated an annual net profit reduction of $302.06 million due to PRRSV compared to $380.82 million in the present study. For the growing phase, the 2013 study estimated a $361.85 million annual loss in net income compared to $819.41 million in the present study. Combined, the financial impact now reaches approximately $1.2 billion per year, emphasizing the increasing PRRSV economic burden on swine producers.

The results from this study indicate that between 2016 and 2020 most growing pigs (> 74 %) were placed PRRSV-negative and less than 60 % of these negative groups stayed negative until market. Across the PRRSV categories for the breeding herds, the least squares mean estimates for LBL, PWM, and LMFY were best in the BH-A category and progressively worsened as the pressure of PRRSV increased, ultimately being at their lowest in the first 16 weeks following an outbreak in PRRSV-unaffected herds, i.e., BH-E category. While this may at first glance seem expected, the AASV classification is solely based on laboratory detection of PRRSV in swine samples; there is therefore no certainty that laboratory outcomes will correlate with productivity outcomes, and by how much. The findings from this study align with a previous study that demonstrated sow productivity improved as breeding herds progressed from an unstable status to stable, provisional negative and negative status, as defined by the AASV classification scheme (Osemeke et al., 2022).

Sow mortality rates also varied by PRRSV status, with BH-A herds having the lowest sow mortality rates of 10.36 % and BH-E herds having the highest sow mortality rate of 12.92 %; the differences in sow death rates between BH categories were not as remarkable as piglet mortality rates. This could be due to the relative resilience of sows to infection compared to the more vulnerable piglets (Klinge et al., 2009), nonetheless, the overall trend indicated that PRRSV instability is associated with poorer sow survival. This finding agrees with a previous study that observed higher mortality in sows that have experienced at least one outbreak of PRRSV (Kikuti et al., 2023).

Unlike the other productivity parameters, the observed statistical differences in the sow culling rates between the PRRSV statuses in the breeding herds lacked a clear directional trend, suggesting that while PRRSV infection is associated with increased sow mortality, the culling rates of sows are driven more by management decisions and other non-PRRSV related factors. This observation is consistent with the previous study by Holtkamp et al., where the culling rate of sows was not observed to be statistically associated with the PRRSV status of the breeding herd (Holtkamp et al., 2013).

For the growing pig herds, groups classified as GH-A had numerically better outcomes for the ADG, FCR, and PPPM, which were not significantly better statistically ($p > 0.05$). Mortality rates, however, showed a clear trend, with GH-A herds having the statistically lowest mortality rate of 7.02 % and GH-C herds the highest at 12.23 %. The mortality difference between GH-A and GH-C groups was statistically significant, aligning with findings from previous studies that demonstrated PRRSV infections in growing pigs contribute significantly to increased mortality and reduced growth rates (Cornelison et al., 2018; Moura et al., 2021). The lack of statistical significance in the productivity differences between the PRRSV statuses for the growing herds may be due in part to the reaggregating of productivity data at the level of the swine companies, resulting in a small sample size of only 9; consequently, reducing the statistical power of the study. Notwithstanding, the best productivity was observed in the groups that stayed PRRSV-negative until market while the worst productivity was observed in groups that were PRRSV-positive at weaning.

The updated estimates demonstrate that despite investments in various areas, PRRSV continues to affect the US swine industry significantly. Metrics such as the number of weaned and marketed pigs, live and carcass weights, and financial losses indicate a notable ongoing burden. Specifically, in the 2013 study, PRRSV was estimated to decrease annual weaned pig numbers by 8.3 million; the recent data shows a reduction of 10.6 million weaned pigs per year. Similarly, while the 2013 study estimated 10 million fewer pigs marketed annually due to PRRSV, recent data show a reduction of 12.6 million marketed pigs, suggesting continued effects at both phases of swine production. Not surprisingly, the estimated losses in live and carcass weights have also increased. The 2013 study reported reductions of 1.46 billion kg in live weight and 1.09 billion kg in carcass weight. Recent data reflect larger losses of 2.09 billion kg in live weight and 1.57 billion kg in carcass weight, indicating sustained challenges in production efficiency.

The 2005, 2013, and current studies estimated that 11.91 %,

45.50 %, and 31.73 %, respectively, of the total losses attributable to PRRSV were from the breeding-herd phase of swine production. Nearly 80 percent of the overall increase in the economic impact of PRRSV in the present study, compared to the 2013 study, was due to losses in the growing-pig herd.

From the 2013 estimate to the current, there has been an increase of about $536 million, with 20.2 % of this increase arising from changes in prices and costs and the size of the US swine herd. The majority (79.8 %) of the increase from the 2013 estimate emanated from changes in the PRRSV status distribution of US breeding sows and the differences in productivity between PRRSV-affected and unaffected herds. This finding is noteworthy as the two factors jointly driving the bulk of the increase in PRRSV cost are largely within the swine practitioner's and producer's control. This further underscores the need for continued innovation in PRRSV prevention and control strategies, more fortification of farm biosecurity, especially in the growing phase of swine production, where we observed the most significant increase in cost.

While scientific insights into the ecology of PRRSV and measures to abate PRRSV may have evolved since the last estimate, the disease continues to impose a significant economic burden on the US swine industry.

This study faced some limitations. Like the previous studies, we relied on the convenience sampling of producers willing to share production records. Additionally, we applied specific selection criteria for the growing herds to ensure accurate PRRSV-status data, including focusing on herds managed as all-in-all-out by site, with known PRRSV status from weaning to when they were sold. While these criteria reduce the likelihood of misclassifying groups of pigs, they may also limit the generalizability of our findings to the broader industry. Additionally, the survey data may have been subject to recall bias.

The analysis was also limited by the availability of estmates of the distribution of herds according to their status for PEDV and Mhp and combinations of all those pathogens and PRRSV. Consequently, the enterprise-budgeting model was not built to analyze the productivity of all combinations of the pathogen statuses, despite the statistical significance of the interaction terms in the analysis of productivity. Only PRRSV main-effects on productivity were modeled, potentially obscuring economically relevant interaction effects. In addition, although variability was reported in the productivity estimates, this variability was not carried forward into the economic calculations, leaving the final cost estimates deterministic. An assessment of the variability of the impact of PRRSV was beyond the scope of this study.

Furthermore, PRRSV's effect on herd productivity has broad economic implications for the national pork supply and pricing. In our findings, we observed that the herd-level productivity losses could lead to fewer pigs marketed annually, potentially affecting the US pork supply by millions of pounds in carcass weight. If PRRSV were eliminated, increased pork supply could lower market prices and enhance US competitiveness in global pork exports. However, the broader economic impact of PRRSV elimination on demand and price dynamics domestically and on exported pork was beyond the scope of this study. Like the 2013 study, our estimates do not capture the unique and often higher costs associated with PRRSV outbreaks in genetic supplier herds, which supply replacement breeding females and boar studs. Although PRRSV outbreaks in these high-health herds are rare, when they do occur, the costs are substantial, encompassing disrupted genetic supply, lost sales, intensive diagnostic testing, and rigorous eradication efforts to regain PRRSV-free status. Thus, while our analysis provides a robust estimate of PRRSV's impact on typical commercial production, the industry-wide cost of PRRSV may be underestimated due to the exclusion of these genetic supplier-related losses. The covid-19 pandemic started within the study period and had multiple adverse effects on the US pork industry (Hayes et al., 2021); while this is expected to have contributed to in some way to the current cost, our methods were not designed to specifically isolate or measure these pandemic-related impacts.

The estimates presented in this study primarily reflect the costs associated with lost productivity due to PRRSV. However, it is important to recognize that the overall economic burden of PRRSV extends far beyond production losses. This includes, but is not limited to, expenses related to veterinary care and diagnostics, biosecurity investments required to mitigate PRRSV entry or spread, as well as the opportunity costs associated with disease management including practices to eliminate the virus from sow herds such as her closures and complete depopulation and repopulation with PRRSV-naïve breeding animals. Social and external effects, such as the impact on the mental well-being of farm owner and staff, the impact of increased carcass disposal on local communities, the lessened environmental wastes from pigs not produced, and similar burdens on the broader agricultural industry, were also outside the scope of this analysis. Future research should aim to provide a comprehensive assessment that incorporates these additional dimensions to fully understand the true socioeconomic impact of PRRSV.

5. Conclusion

The increasing economic impact of PRRSV attributable to suboptimal swine productivity in the US is concerning. Our findings underscore the growing challenges PRRSV poses, reinforcing the critical need for sustained and innovative investments in prevention, control, and elimination.

CRediT authorship contribution statement

Osemeke Onyekachukwu Henry: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Silva Gustavo De-Sousa-E:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Cesar A. Corzo:** Writing – review & editing, Validation, Supervision, Investigation. **Mariana Kikuti:** Writing – review & editing, Visualization, Validation, Resources, Investigation, Formal analysis. **Sarah Vadnais:** Writing – review & editing, Resources, Data curation. **Xiaomei Yue:** Writing – review & editing, Validation, Resources, Investigation, Data curation. **Linhares Daniel C.L.:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Holtkamp Derald J.:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of Competing Interest

The authors of this work declare no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.prevetmed.2025.106627.

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